Amir Taherin
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Professional Summary

I am a PhD candidate working on **efficient and dependable AI systems for edge platforms**. My research spans **generative AI** (large language models and retrieval-augmented generation), **visual perception**, and **physical AI** (robotics and vision-language-action pipelines). I have led multi-student research teams, deployed and characterized AI workloads across three generations of NVIDIA Jetson SoCs and datacenter GPUs, and built reproducible measurement pipelines. My broader goal is to make AI efficient enough to deploy, predictable enough to trust, and robust enough to operate continuously in real-world environments.

Education

PhD in Computer Engineering — **Northeastern University**
Advisors: Profs. David Kaeli and Yanzhi Wang — *2020–Present*

MS in Computer Science — **University of Rochester**
2018–2020

MS in Computer Systems Architecture — **Sharif University of Technology**
Advisor: Prof. Alireza Ejlali — *2014–2016*

BS in Computer Engineering — **K. N. Toosi University of Technology**
2006–2011

Research Experience

NUCAR Laboratory, Northeastern University — **Boston, MA**
Graduate Research Assistant — *2021–Present*

Robotics & Physical AI

- Led a multi-student research team to build an **end-to-end VLA pipeline** for generalist robotic policy learning, maintaining direct collaboration with industry.
- Deployed and benchmarked VLA models (OpenVLA, OpenVLA-oft, SpatialVLA, and our VOTE) on **NVIDIA Jetson AGX Orin** under various power budgets and on **high-performance GPUs** (H100, A100, V100, A6000).
- Upgraded OpenVLA-oft by replacing the LLM backbone with Qwen and Moxin models and redesigning the action head to achieve faster inference on embedded platforms.
- Integrated the full VLA stack with Kinova robotic arms for real-world manipulation experiments.

Generative AI on Edge Systems

- Led a multi-student research team to perform a **system-level characterization of LLM inference** on embedded edge platforms.
- Deployed and benchmarked 13 LLMs (1B-8B parameters) from diverse model families–including LLaMA, Qwen, Gemma, Granite, Mistral, Phi, and Moxin–on **NVIDIA Jetson AGX Xavier, Orin, and Thor** (three consecutive SoC generations).
- Evaluated prompt and instruction-following workloads using the **HuggingFace Transformers** stack and the **IFEval** benchmark suite.
- Executed all models using **Llama.cpp** under multiple **quantization formats** (Q8, Q6, Q4) to study quantization-induced effects on latency, memory behavior, and throughput.
- Analyzed how quantization interacts with **SoC architecture, memory hierarchy, DVFS behavior**, and **GPU scheduling policies** on Xavier (Volta) and Orin (Ampere).
- Designed a unified profiling pipeline capturing **TTFT**, **token latency**, **effective memory bandwidth**, **KV-cache behavior**, **thermal characteristics**, and **GPU/CPU/Memory power** metrics.

- Co-developed **RAGMark**, a stage-level benchmarking framework for retrieval-augmented generation, and **adaptive context compression** for edge RAG.

Efficient Edge Vision

- Developed a lightweight, workload-aware framework for adaptive visual inference that improves energy efficiency and increases object-detection accuracy **without violating real-time constraints** on embedded systems.
- Designed a parallel **Bayesian Optimization** method to balance the trade-off between high-resolution visual inference and runtime efficiency by adaptively adjusting input resolution.
- Implemented **reinforcement learning** baselines for dynamic resolution selection and evaluated their performance relative to the proposed BO-based approach.
- Designed **ALBIREO**, a detector-agnostic, training-free adaptive inference framework for video object detection on edge GPUs (SEC 2026).

Goodwill Laboratory, Northeastern University — **Boston, MA**
Graduate Research Assistant — *2020–2021*

HPC Reliability and Failure Analysis

- Conducted reliability analysis of large-scale **GPU-accelerated supercomputers.**
- Analyzed multi-year failure and repair logs from the Tsubame-2 and Tsubame-3 supercomputers to characterize fault behavior across generations of **multi-GPU compute nodes**.
- Identified systemic trends in GPU, node, and interconnect failures, and examined recovery behavior of large-scale HPC systems.

Systems Group, University of Rochester — **Rochester, NY**
Graduate Research Assistant — *2018–2020*

Energy-Efficient 360° Video Rendering on FPGAs

- Developed an **algorithm-architecture co-designed** system for real-time 360° **AR/VR video rendering** targeting FPGA acceleration.
- Addressed the prohibitive on-chip memory footprint of naive AR/VR rendering pipelines by restructuring the underlying video processing algorithm.
- Implemented the system on Zynq UltraScale+ MPSoC Evaluation Kit, enabling energy-efficient 360° video processing without loss of performance compared to commercial AR/VR rendering systems.

ESRLab, Sharif University of Technology — **Tehran, Iran**
Graduate Research Assistant — *2014–2017*

Reliability-Aware Energy Management in Mixed-Criticality Systems

- Proposed a **reliability-aware energy management** framework for mixed-criticality systems, targeting safe energy reduction in non-safety-critical workloads.
- Designed and evaluated three optimization techniques–**Monotonous-DVFS**, **Stretch**, and a combined DVFS/Stretch method–to exploit slack and degrade low-criticality service levels in a controlled manner.
- Achieved high energy savings with bounded service degradation while preserving system reliability, validated through extensive experiments.

Publications

To Appear

2026: **Hydra: Phase-Aware Workload Characterization of LLM Inference across Edge SoC Generations, Backends, and Quantization Levels**. **Amir Taherin**, Sana Taghipour Anvari, Charles Amante, Yixiao Chen, Ruben Noroian, Zlatan Feric, Nicolas Bohm Agostini, Pu Zhao, José Cano, Bin Ren, Yanzhi Wang, David Kaeli. *IEEE International Symposium on Workload Characterization (IISWC)*, 2026. To appear.

2026: **RAGMark: A Comprehensive Framework for Benchmarking Retrieval-Augmented Generation Systems**. Zlatan Feric, **Amir Taherin**, Bin Ren, Yanzhi Wang, Jennifer Dy, David Kaeli. *IEEE International Symposium on Workload Characterization (IISWC)*, 2026. To appear.

2026: **ALBIREO: Adaptive, Energy-Efficient Inference Framework for Video Object Detection on the Edge**. **Amir Taherin**, José Cano, Bin Ren, Yanzhi Wang, David Kaeli. *ACM/IEEE Symposium on Edge Computing (SEC)*, 2026. To appear.

2026: **From Retrieved Context to Runtime Control: Adaptive Compression for Edge-based RAG**. Zlatan Feric*, **Amir Taherin***, Yanzhi Wang, David Kaeli. *Proceedings of the ACM AI Leadership Summit*, 2026. To appear. *Equal contribution.

Preprints

2026: **Human Cognition in Machines: A Unified Perspective of World Models**. Timothy Rupprecht*, Pu Zhao*, **Amir Taherin***, Arash Akbari*, Arman Akbari*, Yumei He*, Tooba Imtiaz*, Sean Duffy, Juyi Lin, Yixiao Chen, Rahul Chowdhury, Enfu Nan, Yixin Shen, Yifan Cao, Haochen Zeng, Weiwei Chen, Geng Yuan, Jennifer Dy, Sarah Ostadabbas, Xuan Zhang, David Kaeli, Edmund Yeh, Yanzhi Wang. arXiv:2604.16592, 2026. *Equal contribution.

2025: **VOTE: Vision-Language-Action Optimization with Trajectory Ensemble Voting**. Juyi Lin, **Amir Taherin**, Arash Akbari, Arman Akbari, Lei Lu, Guangyu Chen, Taskin Padir, Xiaomeng Yang, Weiwei Chen, Yiqian Li, Xue Lin, David Kaeli, Pu Zhao, Yanzhi Wang, arXiv:2507.05116, 2025.

Journal Papers

2018: **Reliability-Aware Energy Management in Mixed-Criticality Systems**. **Amir Taherin**, Mohammad Salehi, Alireza Ejlali. *IEEE Transactions on Sustainable Computing*, doi:10.1109/TSUSC.2018.2801123, 2018.

Conference Papers

2026: **Cross-Platform Scaling of Vision-Language-Action Models from Edge to Cloud GPUs**. **Amir Taherin**, Juyi Lin, Arash Akbari, Arman Akbari, Pu Zhao, Weiwei Chen, David Kaeli, Yanzhi Wang. *Proceedings of the Great Lakes Symposium on VLSI (GLSVLSI '26)*, doi:10.1145/3787109.3816400, 2026.

2021: **Examining Failures and Repairs on Supercomputers with Multi-GPU Compute Nodes**. **Amir Taherin**, Tirthak Patel, Giorgis Georgakoudis, Ignacio Laguna, and Devesh Tiwari. *In The 51st Annual IEEE/IFIP International Conference on Dependable Systems and Networks*, doi:10.1109/DSN48987.2021.00043, 2021.

2020: **Energy-Efficient 360-Degree Video Rendering on FPGA via Algorithm-Architecture Co-Design**. Qiuyue Sun, **Amir Taherin**, Yawo Siatitse, and Yuhao Zhu. *In The 2020 ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA '20). Association for Computing Machinery*, doi:10.1145/3373087.3375317, 2020.

2015: **Stretch: Exploiting Service Level Degradation for Energy Management in Mixed-Criticality Systems**. **Amir Taherin**, Mohammad Salehi, Alireza Ejlali. *The CSI Symposium on Real-Time and Embedded Systems and Technologies (RTEST)*, doi:10.1109/RTEST.2015.7369846, 2015.

Technical Skills

Programming: C/C++ (OpenMP, MPI, pthreads), Python, CUDA, Verilog, Bash, MATLAB, x86/ARM Assembly

ML/AI: PyTorch, TensorRT-LLM, ONNX, Llama.cpp

Dev. Boards: NVIDIA Jetson AGX Thor, Jetson AGX Orin, Jetson AGX Xavier, Zynq UltraScale+ MPSoC ZCU104 Evaluation Kit

CAD Tools: Synopsys (*Design Compiler, HSPICE, PrimePower, Platform Architect*), Cadence (*Virtuoso, SoC Encounter*), Mentor Graphics (*ModelSim*), Xilinx (*ISE Design Suite, Vivado HLS, SDSoC*), Simulink

Typesetting: LaTeX, TeX, Markdown

Research Interests

- Efficient Generative AI Inference (LLMs, RAG)
- Computer Architecture, SoC, and GPU Systems
- Edge Computing and AI Acceleration
- Physical AI, Robotics, and VLA Models

Teaching Experience

Teaching Assistant, High Performance Computing — **Northeastern University**
Course Instructor: Prof. David Kaeli — *Fall 2022*

Teaching Assistant, Parallel and Distributed Computing — **University of Rochester**
Course Instructor: Prof. Sandhya Dwarkadas — *Spring 2020*

Teaching Assistant, Programming Languages Design and Implementation — **University of Rochester**
Course Instructor: Prof. Michael L. Scott — *Fall 2019*

Teaching Assistant, Computer Organization — **University of Rochester**
Course Instructor: Prof. Yuhao Zhu — *Spring 2019*

Teaching Assistant, Embedded Systems Design — **Sharif University of Technology**
Course Instructor: Prof. Alireza Ejlali — *Spring 2016*

Teaching Assistant, Logic Design — **Sharif University of Technology**
Course Instructor: Prof. Shaahin Hessabi — *Spring 2016*

Teaching Assistant, Advanced Logic Design — **Sharif University of Technology**
Course Instructor: Prof. Alireza Ejlali — *Spring 2015*

Honors & Awards

2026: **Future Leaders of AI Doctoral Consortium**, selected participant, ACM AI Leadership Summit, Atlanta.

2022: **Quantum Excellence** in Quantum Simulation from IBM Qiskit Global Summer School

2021: **Quantum Excellence** in Quantum Machine Learning from IBM Qiskit Global Summer School

2016: **Ranked 3rd** in cumulative GPA among all students of computer architecture (41 students), Sharif University of Technology, Tehran, Iran.

2015: **National Talent Award** for exceptional GPA from Sharif University of Technology, Tehran, Iran.

Academic Services

Solicited Reviewer: IEEE TPAMI (2026), IEEE TCC (2026), ACM CSUR (2026), ACM TACO (2026), IEEE TETC (2018), RTEST (2017, 2015)

Artifacts: Released the **Hydra** characterization framework and ∼107K-record measurement corpus (GitHub; Zenodo doi:10.5281/zenodo.21844843)

Languages

◦ **English**: Full professional proficiency (TOEFL 115/120) ◦ **Persian**: Native